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# npm vs npx

Choosing the right tool for the job



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# npm

**npm** (*Node Package Manager*) is a tool for managing JavaScript packages.

It allows you to:

1. **Install**, **update**, and **manage** dependencies for your project.
2. Add **global** or local packages to your environment.
3. Maintain a **package.json** file to track your project's dependencies.

Use Case: *npm install react*



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# npx

**npx** (*Node Package eXecute*) is a utility for running Node.js packages directly, without globally installing them.

It allows you to:

1. Add local packages to your environment, without globally installing them.

Use Case: *npx create-react-app my-app*



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# Key Differences

## 1. Installation

- **npm** installs packages globally or locally.
- **npx** doesn't install packages; it runs them temporarily.

## 2. Temporary Usage

- Use **npx** to avoid cluttering your global namespace with unnecessary tools.
  - Instead of **npm install -g eslint**,
  - Use **npx eslint .** to run ESLint.



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# Key Differences

## 3. Version Handling:

- **npm** allows you to run a specific package version without installing it.
  - **npx webpack@4.0.0** -> runs Webpack of version 4.0.0.

## 4. Convenience

- **npm** is essential for long-term dependency management.
- **npx** is perfect for one-time tasks like creating a new project or running a CLI command.



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# When to Use Each

## Use **npm** when:

- You need to add a **dependency** to your project.
- You want to **manage** and **maintain** packages in **package.json**.

## Use **npx** when:

- You need to **run a tool** or **package** temporarily.
- You want to **test a package** before installing it.



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# In Summary

Use **npm** for managing dependencies

and

Use **npx** for quick package execution  
without installation.



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